EE3032 Introduction to VLSI—Midterm

1. (30 points) Complete the following problems briefly.

   (a) During the process of VLSI manufacturing, the chemical-mechanical polishing (CMP) is performed while a layer of material is grown. What is the purpose of CMP? (3 points)

   (b) The layer density rule should be met when you design a nano-scale IC. Why? (3 points)

   (c) What is the antenna effect? (3 points)

   (d) Draw a CMOS inverter and label the drain and source of the PMOS and NMOS transistors. (2 points)

   (e) Define the low and high noise margins of a CMOS inverter using the voltage transfer curve of the inverter. (3 points)

   (f) List two possible approaches for alleviating the effect of crosstalk on two neighboring parallel interconnects. (4 points)

   (g) List three major components of the load capacitance of a CMOS logic gate in a CMOS circuit. (3 points)

   (h) How do you size the N-type and P-type MOS transistors of an inverter such that the output voltage of the inverter is $V_{dd}/2$ when the input voltage of the inverter is $V_{dd}/2$? (3 points)

   (i) List two types of FinFET structures. (3 points)

   (j) The delay of a *wire* interconnection can be estimated by kx^2, where k is a constant. What approach can be used to reduce the delay of interconnection? Explain your answer. (3 points)

approach can be used to reduce the delay

2. (15 points) Figure 1 shows a transistor-level schematic realizing a Boolean function f.

(a) Assume that the sizes of all PMOS transistors and NMOS transistors are the same and the turn-on resistance of PMOS and NMOS transistors is R_p and R_n, respectively. Also, parasitic capacitances of nodes 1, 2, 3, and 4 are C_1, C_2, C_3, and C_4, respectively, and $C_1 = C_2 = C_3 = C_4 = C$. If functional inputs (XYZ) of the circuit are {010, 100, 110, 000, 101, 111}, calculate the worst-case rise time and fall time of the circuit using Elmore RC delay model. (10 points)

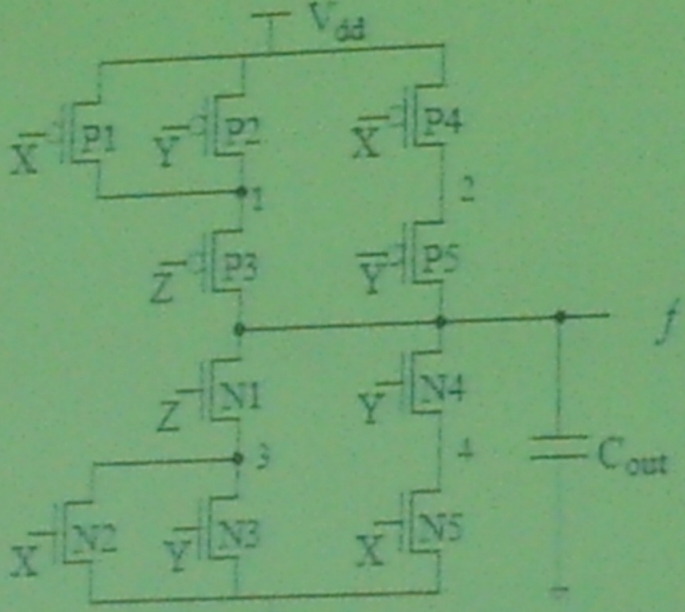


Figure 1: Transistor-level schematic

(b) Assume that the voltage swings at the nodes 1, 2, 3, and 4 are V_1, V_2, V_3, and V_4. Also, the switching activities of the nodes 1, 2, 3, and 4 are α_1, α_2, α_3, and α_4. If the logic is operated at the frequency f. Express the internal cell power of the logic gate. (5 points)

3. (10 points) Figure 2(a) depicts a CMOS logic circuit in which the sizes of NMOS and PMOS for implementing each gate are the same. Assume that a dual-V_t process can be used to implement the CMOS circuit. Draw the transistor schematic diagram of the CMOS circuit in terms of the symbols shown in Fig. 2(b) such that the leakage power of the CMOS circuit is minimal without sacrificing the speed of the CMOS circuit.

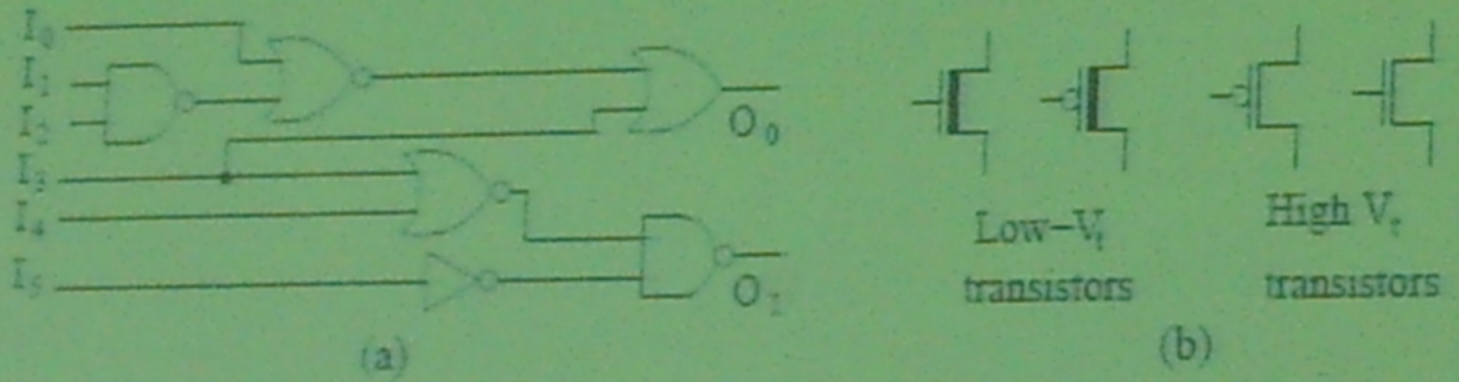


Figure 2: (a) A CMOS circuit. (b) Symbols of high-V_t and low-V_t transistors.

4. (10 points) Figure 3 shows a pass transistor logic circuit. Assume that the supply voltage $V_{dd} = 1.8V$ and the threshold voltage of the NMOS transistor is $0.4V$.

(a) Draw the corresponding waveforms at the output Y and Z if the signal waveform shown in the leftmost figure is applied to the input X. Note that you should label the smallest and largest values of voltage in the waveforms. (4 points)

(b) Assume that the body effect of the transistors can be neglected. Also, the signal waveform shown in the leftmost figure is applied to the input X.

i. Among the transistors $m1$, $m2$, $m3$, $m4$, $m5$, and $m6$, identify the transistor(s) which can be operated in saturation and linear regions, and identify the transistor(s) which can be operated in linear region only. Explain your answers. (6 points)

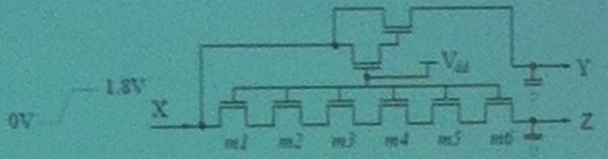


Figure 3: A pass transistor logic circuit.

5. (10 points) Figure 4 depicts an inverter chain driving a capacitive load C_L. Assume that the three inverters have the same size and every inverter has a minimum-size NMOS and the ratio of PMOS to NMOS is α. Also, the t_r and t_f of an inverter driving a load C_L can be estimated by $t_r = R_p \times C_L$ and $t_f = R_n \times C_L$, where R_p and R_n are the turn-on resistance of the PMOS and NMOS of the inverter. Let $\mu_n = 3\mu_p$ and the turn-on resistance and gate capacitance of the minimum-size NMOS are R and C, respectively. Derive the value of α such that the $T = 8RC$ and the average switching power constituted by the first two inverters is minimal.

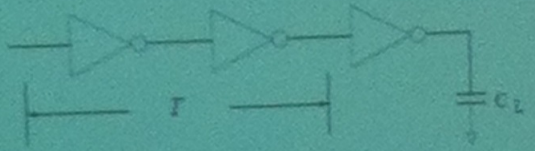


Figure 4: An inverter chain.

6. (10 points) Realize the following Boolean functions using static CMOS logic. You should show the transistor diagram.

   (a) $f(w,x,y,z) = \overline{w(x+y+z)}$ (5 points)

   (b) $f(x,y) = \bar{x}y + x\bar{y}$ (5 points)

   Suppose that only true literals are available. Your implementation should use the minimal number of transistors.

7. (15 points) A full adder is a device that accepts three inputs x, y, and c_{in}, and produces two outputs s_{out} and c_{out}, where x, y, c_{in}, s_{out}, and c_{out} denote the addend, the augend, carry input, sum, and carry output, respectively.

   (a) Derive the truth table for this full adder. (5 points)

   (b) From the truth table, derive the s_{out} and c_{out} Boolean function. (5 points)

   (c) Realize it using CMOS logic circuit. (5 points)